

Highlights

Lesion-Grounded Ordinal Grading with Clinical-Rule Consistency and Conformal Referral for Diabetic Retinopathy: Joint Classification, Lesion Segmentation and Lesion Detection on the DDR Dataset

First Author, Second Author

- TODO

Lesion-Grounded Ordinal Grading with Clinical-Rule Consistency and Conformal Referral for Diabetic Retinopathy: Joint Classification, Lesion Segmentation and Lesion Detection on the DDR Dataset

First Author^{a,*}, Second Author^a

^a*Department / Institution, City, Country*

Abstract

TODO

Keywords: diabetic retinopathy, ordinal classification, lesion segmentation, lesion detection, clinical rule consistency, conformal prediction, fundus photography

1. Introduction

Diabetic retinopathy (DR) is a microvascular complication of diabetes mellitus and remains a leading cause of preventable vision impairment among working-age adults worldwide (Saeedi et al., 2019; Teo et al., 2021). A recent meta-analysis estimated that about one in five people living with diabetes (22.3%) has some degree of retinopathy, and projected that the absolute burden will keep increasing through the middle of this century as the prevalence of diabetes rises (Teo et al., 2021). Vision loss from DR is largely avoidable when disease is found early and treated on time, which is why periodic fundus photography is recommended for every person with diabetes. The bottleneck is not the camera but the reader: grading millions of photographs per year requires trained graders and ophthalmologists who are scarce in exactly those regions where diabetes is growing fastest. This has made automated DR

*Corresponding author. (Author names, affiliations and e-mail are placeholders to be completed before submission.)

Email address: `corresponding.author@example.org` (First Author)

screening one of the most extensively studied applications of deep learning in medical imaging, from the first large-scale validations (Gulshan et al., 2016; Ting et al., 2017) to prospective trials of autonomous systems in primary care (Abràmoff et al., 2018).

Clinically, DR is graded from a small set of visible lesions. Microaneurysms (MA), intraretinal haemorrhages (HE), hard exudates (EX) and soft exudates or cotton-wool spots (SE) are the hallmarks of non-proliferative disease, and neovascularisation marks the proliferative stage. The International Clinical Diabetic Retinopathy (ICDR) severity scale (Wilkinson et al., 2003), a simplification of the ETDRS classification (Early Treatment Diabetic Retinopathy Study Research Group, 1991), defines five grades — no DR, mild, moderate, severe non-proliferative DR and proliferative DR — by stating which lesions must be present and, for the severe stage, how widely they must be distributed (the well-known “4–2–1” rule: haemorrhages in four quadrants, venous beading in two, or intraretinal microvascular abnormalities in one). Three machine-learning tasks therefore describe the same clinical reasoning at different granularity: *grading* (image level, five ordered classes), *lesion segmentation* (pixel level) and *lesion detection* (instance level).

In the literature, however, these three tasks are almost always solved separately. Grading networks are trained on image-level labels and are asked to rediscover lesions implicitly, which makes them data-hungry and hard to explain. Segmentation networks are trained on a few hundred pixel-annotated images and are scored by overlap measures that say little about whether a clinician could use the output. Detection networks are borrowed from natural-image object detection and struggle with objects that occupy a handful of pixels. Multi-task models that share an encoder do exist (Foo et al., 2020; Playout et al., 2019; Zhou et al., 2019), as do lesion-aware grading models (Huang et al., 2021; Sun et al., 2021; Wang et al., 2017), but in most of them the link between lesion evidence and the final grade is learned implicitly through shared weights. Nothing prevents such a model from predicting *severe* DR for an image in which it finds no haemorrhage at all, and the practitioner has no principled way to know when to distrust the output.

Gaps addressed in this work.. We identify four specific gaps. **(G1) Lesion-grade inconsistency.** Grade predictions are not required to agree with the lesion evidence produced by the same system, so internally contradictory outputs are possible and are rarely measured. **(G2) Ordinal structure and label noise.** DR grades are ordered, and inter-grader disagreement

concentrates on adjacent grades, yet cross-entropy over nominal classes ignores both facts. **(G3) Small-lesion sensitivity.** Microaneurysms are a few pixels wide in a megapixel image; losses and architectures that are fine for organs are blind to them. **(G4) Absence of principled referral.** Most systems output a single hard label with a miscalibrated confidence; very few provide a finite-sample statement about how often the clinically true grade is contained in the output, which is what a referral workflow needs.

Our approach.. We propose LesionRule, a lesion-grounded grading framework built on top of a lesion-segmentation network. The segmentation network produces four lesion-probability maps. A differentiable, connected-component-free *lesion-evidence bottleneck* converts those maps into soft per-zone lesion statistics (counts at several confidence levels, number of local peaks and maximum probability for each of four lesion types in each of six retinal zones). Each (lesion, zone) statistic becomes a token; a small transformer lets a grade query attend jointly to a global image embedding and to these 24 evidence tokens, so that the grade is computed *from* the lesions the same network sees. A *clinical-rule consistency loss* encodes ICDR-style necessary conditions (“grade $\geq$ severe requires haemorrhage in all four quadrants or heavy soft exudates”) as soft logic with a product t-norm, with thresholds calibrated on the training labels, and penalises predictions that exceed what the lesion evidence permits. The grade head is *ordinal* with Gaussian-smoothed soft targets that reflect adjacent-grade disagreement, and the resulting class probabilities are temperature-scaled and wrapped by *split-conformal prediction* to give referral sets with a finite-sample coverage guarantee. On the lesion side, a size-aware focal-Tversky objective with a full-resolution shallow branch is used to keep microaneurysms visible, and the very same probability maps are converted to detection boxes by connected-component proposals, so that segmentation and detection share one set of weights.

Experimental scope and honesty about compute.. All experiments reported here were produced on a single four-core CPU workstation without a GPU (Section 5). This is a deliberate design constraint rather than an incidental one: it forces the study to be explicit about what is learned end-to-end (the lesion segmenter) and what is learned on top of frozen representations (the grading heads), and it makes all reported numbers reproducible without specialised hardware. It also means that absolute grading accuracy is below what an end-to-end fine-tuned, high-resolution, GPU-trained model could

reach, and we do not claim state-of-the-art accuracy on the DDR benchmark. What we do claim, and test with controlled ablations, multiple seeds and confidence intervals, is that *given the same frozen image embedding*, grounding the grade in lesion evidence and enforcing rule consistency yields measurable gains in agreement, internal consistency, calibration and referral quality over the standard alternatives.

Contributions..

1. We formulate DR grading, lesion segmentation and lesion detection as a single coupled pipeline in which the grade is a function of differentiable, zone-wise lesion statistics (Section 4.5), and we make this coupling explicit with a token-based evidence head.
2. We introduce a soft-logic *clinical-rule consistency* regulariser with learnable, label-calibrated thresholds and a matching *rule-violation rate* metric that quantifies internal contradictions between lesion evidence and predicted grade (Section 4.8).
3. We combine ordinal cumulative-link learning with Gaussian-smoothed soft targets and split-conformal prediction to obtain calibrated grade distributions and referral sets with finite-sample coverage (Sections 4.7 and 4.9).
4. We provide a size-aware lesion segmentation recipe and a segmentation-derived detection protocol with FROC analysis for lesions that are only a few pixels wide, evaluated on the full DDR lesion subset (Section 4.3).
5. We report a comprehensive and fully reproducible empirical study on the DDR dataset (Li et al., 2019) — 13,673 fundus images, 757 of them with pixel masks and boxes — including loss and preprocessing ablations, component ablations over five seeds, calibration and conformal analyses, and a zero-shot external test on the Gaussian-filtered APTOS-derived Kaggle set (Asia Pacific Tele-Ophthalmology Society, 2019; Rath, 2020) (Section ??). All code and cached lesion maps are released.

Organisation.. Section 2 reviews related work and positions LesionRule against it. Section 3 describes the datasets and preprocessing. Section 4 details the method, with mathematical formulation and algorithms. Section 5 gives the experimental protocol. Section ?? presents results, and Section ?? discusses their clinical meaning, limitations and ethical aspects. Section ?? concludes.

2. Related work

2.1. Deep learning for DR grading

The modern era of automated DR screening started with convolutional networks trained on large collections of graded photographs (Gulshan et al., 2016), followed by multi-ethnic validations (Ting et al., 2017) and regulatory-grade prospective studies (Abràmoff et al., 2018). Subsequent research on public data has largely consisted of backbone and training-recipe improvements. Residual (He et al., 2016), compound-scaled (Tan and Le, 2019), modernised convolutional (Liu et al., 2022) and transformer backbones (Dosovitskiy et al., 2021; Liu et al., 2021) have all been applied to grading, typically with high-resolution inputs, heavy augmentation and ordinal or regression objectives. Because retinal images are structurally homogeneous and labelled data are scarce compared with natural images, large self-supervised retinal foundation models such as RETFound (Zhou et al., 2023) have recently shown that domain-specific pre-training transfers well across diseases and datasets. Our framework is complementary to this trend: it accepts any image embedding (here a frozen ImageNet ConvNeXt-T (Liu et al., 2022)) and adds lesion-grounded evidence and consistency on top.

A second line of work tackles the specific difficulties of DR grading. The class distribution is heavily skewed towards no-DR and moderate DR, and severe DR is rare. Category-attention modules (He et al., 2021) were proposed to counter this imbalance and were evaluated on DDR. Joint grading of DR and diabetic macular oedema exploits the strong relationship between the two diseases (Li et al., 2020). Noise-aware and attention-based fusion frameworks (Lin et al., 2018) address low-quality inputs. Common to these methods is that they supervise only the grade; the lesions that justify the grade are left implicit.

2.2. Lesion-aware grading and weak lesion localisation

Because lesions are the clinical basis of grading, a natural idea is to give the grader access to lesion information. Zoom-in-Net (Wang et al., 2017) mines suspicious regions with an attention branch trained only from image-level labels and re-examines them at higher resolution. Image-mining approaches (Quellec et al., 2017) derive lesion heatmaps from a trained grading network. Two-stage methods first detect lesions and then feed lesion counts or maps to a grader (Yang et al., 2017). Lesion-aware transformers (Sun

et al., 2021) cast lesion detection and grading in a joint transformer framework, and lesion-based contrastive learning (Huang et al., 2021) uses lesion patches to learn grading-relevant features. These works demonstrate that lesion information improves grading and interpretability. However, to our knowledge, they do not (i) make the grade a function of *zone-resolved* lesion statistics with explicit tokens that can be inspected, (ii) check that the final grade is *consistent* with the lesion evidence according to clinical-rule templates, or (iii) report a rule-violation measure, calibration and conformal coverage jointly. These are the aspects on which LesionRule differs.

2.3. Retinal lesion segmentation

Lesion segmentation in fundus images has been driven by encoder–decoder networks descending from U-Net (Ronneberger et al., 2015), including nested skip connections (Zhou et al., 2020), attention gates (Oktay et al., 2018) and atrous multi-scale decoders (Chen et al., 2018). Public pixel-level resources include IDRiD (Porwal et al., 2020), e-optha (Decencière et al., 2013), the larger FGADR collection (Zhou et al., 2021) and the DDR dataset used here (Li et al., 2019). The consistent finding across these benchmarks is that large, high-contrast lesions (hard exudates, large haemorrhages) are segmented reasonably well while microaneurysms remain difficult: they are small, low-contrast, often indistinguishable from vessel fragments, and their annotation is noisy. Overlap measures such as Dice are dominated by large lesions, whereas class-imbalance-robust measures such as the area under the precision–recall curve (AUPR) are more informative for rare, small objects (Saito and Rehmsmeier, 2015). Loss design is therefore important: Tversky-type losses (Salehi et al., 2017) and their focal variant (Abraham and Khan, 2019) weight false negatives more than false positives and are a standard remedy for tiny foreground classes, while focal modulation (Lin et al., 2017) down-weights easy background pixels. We adopt this family and add an explicit full-resolution branch, and we quantify the benefit through an ablation against binary cross-entropy and Dice-based objectives.

2.4. Lesion detection

Detection of retinal lesions has mostly reused generic object detectors (Ren et al., 2017; Tian et al., 2019). These architectures were designed for objects that span tens to hundreds of pixels, with anchors and strides that are poorly matched to lesions that may span only a few pixels, and they require many training boxes, while the DDR detection subset has only 383

training images. Detection from segmentation outputs is an attractive alternative when pixel masks are available: instances are obtained by connected-component analysis and ranked by their mean probability, and a single network serves both tasks. This is the route we take. Following common practice for tiny-object detection, we report average precision at relaxed and standard overlap thresholds (Padilla et al., 2021) together with lesion-level free-response ROC (FROC) sensitivity at fixed numbers of false positives per image, which is the quantity that matters for screening workflows.

2.5. *Multi-task learning and mixed supervision*

Multi-task models for DR combine grading with lesion segmentation to exploit complementary supervision (Foo et al., 2020), learn lesion segmentation from image-level labels with weak supervision (Playout et al., 2019), or jointly train segmentation and classification in a semi-supervised, collaborative fashion (Zhou et al., 2019). Attention-based multiple-instance learning (Ilse et al., 2018) provides a general mechanism to localise image-level evidence. In all these designs, sharing is realised through common weights and a summed loss. Our contribution to this line is to make the dependence of the grade on the lesion maps *structural* rather than implicit, and to attach logic-based constraints to that dependence.

2.6. *Ordinal learning, label noise and knowledge-based constraints*

DR grades are ordered, so treating them as nominal categories discards information. Ordinal formulations convert a K -class problem to $K-1$ binary cumulative questions (Niu et al., 2016), with rank-consistent variants that guarantee monotone cumulative probabilities (Cao et al., 2020; Shi et al., 2023). Inter-grader disagreement in DR concentrates on adjacent grades, which suggests smoothing the target distribution along the grade axis; we adopt a discretised Gaussian target on the cumulative scale. Injecting logical knowledge into neural networks has a long history: logic rules can regularise network outputs through teacher-student distillation (Hu et al., 2016), semantic-based regularisation with t-norm relaxations (Diligenti et al., 2017), or semantic loss functions (Xu et al., 2018). We use the t-norm relaxation, because it is differentiable, requires no extra labels and gives an interpretable per-rule score. The ICDR/ETDRS grading definitions (Early Treatment Diabetic Retinopathy Study Research Group, 1991; Wilkinson et al., 2003) supply the templates. To our knowledge such rule templates have not previously

been combined with zone-resolved lesion statistics from the same network and an explicit violation-rate metric in DR grading.

2.7. Calibration and conformal prediction

Modern networks are over-confident (Guo et al., 2017), and for screening the practically important question is what the probability that the true grade lies in the set of grades the system proposes is. Split-conformal prediction (Angelopoulos and Bates, 2023; Vovk et al., 2005) converts any probabilistic classifier into a set-valued predictor with finite-sample marginal coverage under exchangeability, without retraining. Monte-Carlo dropout (Gal and Ghahramani, 2016) and deep ensembles estimate epistemic uncertainty but offer no coverage guarantee. We combine temperature scaling for calibration with split-conformal sets, define a clinically motivated referral rule on the resulting sets, and report coverage, set size and referral errors.

2.8. Positioning

Table 1 summarises how the methods discussed above differ in design. The comparison is qualitative: reported numbers of other works are obtained under different splits, resolutions, backbones and compute budgets, and we have not re-run those systems; therefore we make no numerical claims against them and restrict quantitative comparisons to controlled baselines trained under the same protocol (Section 5).

Table 1: Design-level comparison with representative prior work (qualitative; ✓ = addressed explicitly, ◦ = partially or implicitly, − = not addressed). Entries reflect our reading of the cited papers’ main design and may omit secondary features.

Method	Grade	Seg.	Det.	Lesion→grade link	Zone-wise evidence	Rule consist.	Ordinal / n
Zoom-in-Net (Wang et al., 2017)	✓	−	◦	◦	−	−	−
Two-stage detect. + grade (Yang et al., 2017)	✓	−	✓	✓	−	−	−
Foo et al. (Foo et al., 2020)	✓	✓	−	◦	−	−	−
Playout et al. (Play-out et al., 2019)	−	✓	−	−	−	−	−
CANet (Li et al., 2020)	✓	−	−	−	−	−	−
CABNet (He et al., 2021)	✓	−	−	−	−	−	◦
Lesion-aware transf. (Sun et al., 2021)	✓	−	✓	✓	−	−	−
Lesion-based contrastive (Huang et al., 2021)	✓	−	−	◦	−	−	−
LesionRule (this work)	✓	✓	✓	✓	✓	✓	✓

3. Materials

3.1. The DDR dataset

All primary experiments use the DDR dataset (also distributed as OIA-DDR) (Li et al., 2019), a general-purpose collection of colour fundus photographs designed for three tasks: DR grading, lesion segmentation and lesion detection. According to the dataset authors, the photographs were collected from many hospitals and acquired with a variety of fundus cameras, which gives the data heterogeneous colour, illumination and field-of-view characteristics (Fig. 1). We downloaded the complete archive (ten split zip volumes, 15.1 GB) and verified its structure and counts directly from the files: the

grading task contains 13 673 images split into 6835 training, 2733 validation and 4105 test images; the lesion subset contains 757 images (383 / 149 / 225) with four-class pixel-level masks and bounding boxes in Pascal-VOC XML format. The dataset is released under CC BY-NC-SA 4.0, and we use it for non-commercial research with attribution as required.

Grading labels. Each image carries one of six labels: grades 0 (no DR), 1 (mild), 2 (moderate), 3 (severe), 4 (proliferative DR) and 5 (ungradable owing to poor quality). The label distribution is strongly imbalanced (Table 2, Fig. 3a): 46% of training images are grade 0 and 33% grade 2, whereas severe DR (grade 3) accounts for only 118 training images (1.7%). The distribution is nearly identical across the three splits, which suggests stratified assignment. Throughout the paper we use two protocols. The *five-class protocol* (main) restricts grading to gradable images (grades 0–4: 6260 / 2503 / 3759 train / validation / test images), because ordinal reasoning is meaningless for an ungradable photograph; image quality is handled by a separate gate (Section 4.10). The *six-class protocol* adds the ungradable class as a nominal outcome and is reported for completeness.

Lesion annotations. For the 757 lesion images, four lesion types are annotated at pixel level: microaneurysms (MA), haemorrhages (HE), hard exudates (EX) and soft exudates (SE). The annotation is dense (Fig. 2): after removing duplicates, the training subset contains 24 671 annotated boxes, of which 13 419 are exudates, 5783 microaneurysms, 4739 haemorrhages and 730 soft exudates (the raw training XML files list 58 200 boxes, because many boxes appear more than once; the validation and test files contain no duplicates). Lesion size spans two orders of magnitude (Fig. 3c): when sizes are rescaled to the nominal ≈ 1950 -pixel native resolution, the median longer box side is about 10 px for microaneurysms and exudates, 26 px for haemorrhages and 47 px for soft exudates, and the 10th percentile is only 3.6–4.7 px for MA, EX and SE. In the resolution used by most deep networks (512 px or lower), a typical microaneurysm therefore occupies one to three pixels. The detection annotations are derived from the same lesions as the masks. Table 2 also lists the number of lesion images in which each class occurs.

3.2. Gaussian-filtered APTOS-derived Kaggle set (external test)

For an external test of the grading pipeline we use the publicly available “Diabetic Retinopathy 224×224 Gaussian filtered” Kaggle dataset (Rath,

Table 2: DDR dataset statistics as verified from the downloaded archive. Top: grading labels per split (G0–G4 = ICDR grades, UG = ungradable). Bottom: lesion subset (unique boxes after duplicate removal; number of images containing each lesion class in parentheses).

Split		G0	G1	G2	G3	G4	UG	Total
Train		3,133	315	2,238	118	456	575	6,835
Validation		1,253	126	895	47	182	230	2,733
Test		1,880	189	1,344	71	275	346	4,105
All		6,266	630	4,477	236	913	1,151	13,673

Lesion split	Images	MA	HE	EX	SE
Train	383	5,783 (314)	4,739 (294)	13,419 (245)	730 (111)
Validation	149	2,556 (132)	1,342 (113)	1,920 (70)	349 (86)
Test	225	2,041 (124)	6,457 (194)	8,320 (171)	214 (42)

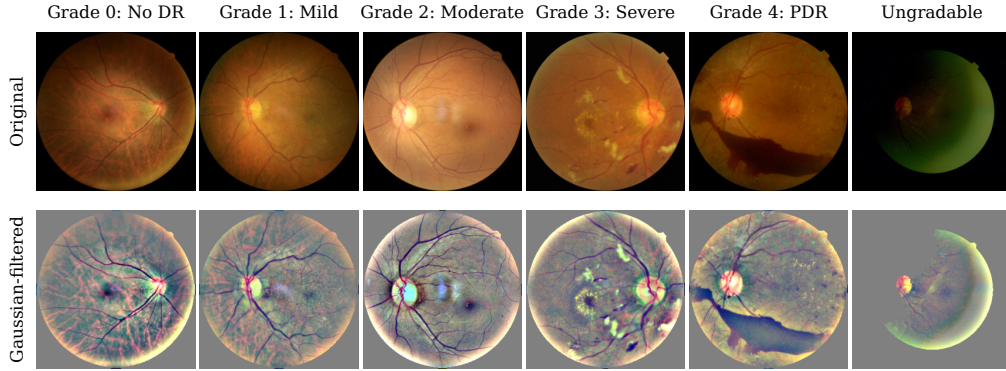


Figure 1: Sample DDR fundus images for each grade (top row: cropped original; bottom row: after Gaussian-filter preprocessing). Samples are drawn at random from the training split with a fixed seed, subject to the field of view filling the frame; the last column shows an ungradable image. The preprocessing equalises illumination and exposes vessels and lesions, but amplifies border artefacts.

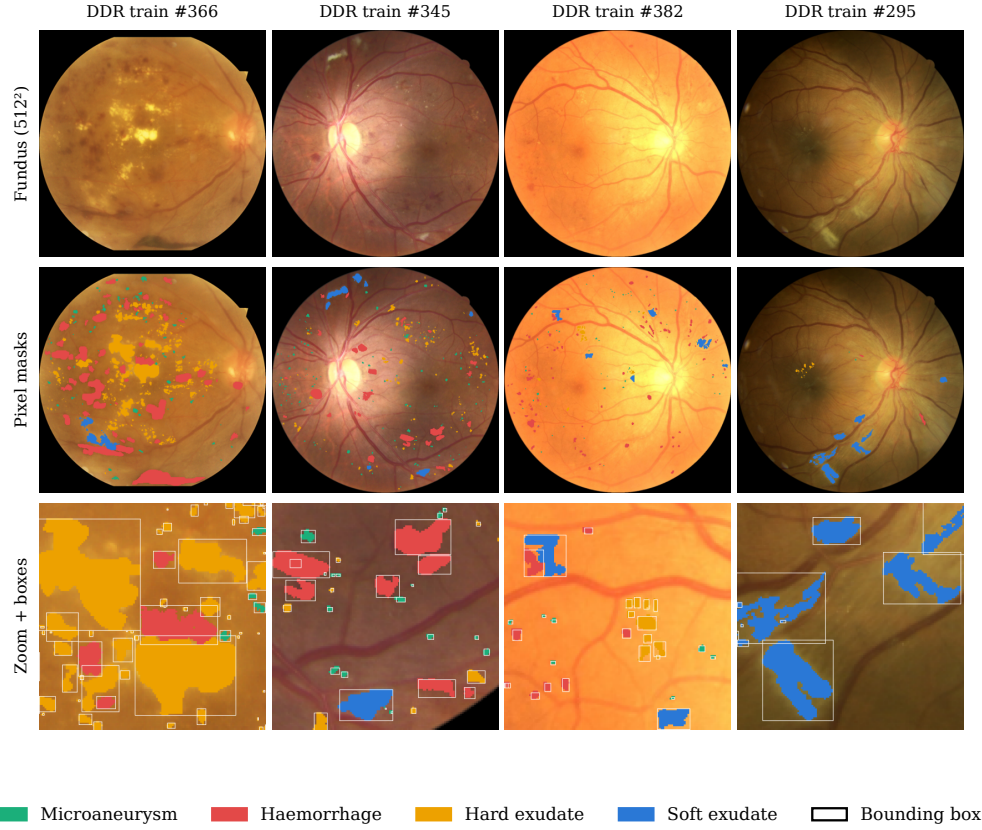


Figure 2: Lesion annotations in four DDR training images that contain several lesion types. Top: fundus image at 512×512 . Middle: ground-truth pixel masks (MA green, HE red, EX yellow, SE blue). Bottom: densest 128×128 window magnified three times with the Pascal-VOC bounding boxes (white). Note the very different sizes and densities of lesions between images.

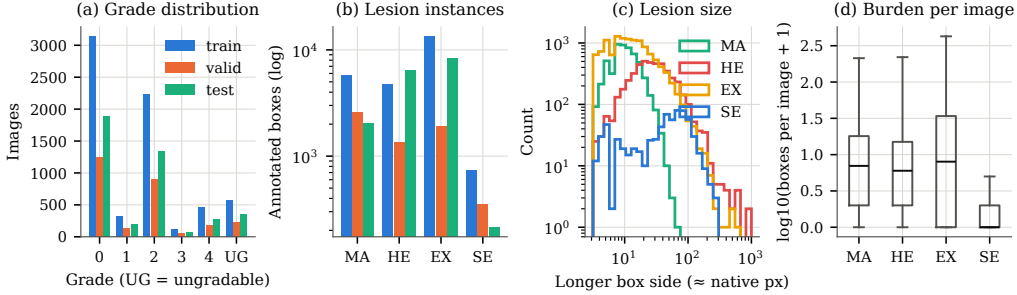


Figure 3: Dataset statistics. (a) Grade distribution per split. (b) Number of annotated lesion boxes per class and split (log scale). (c) Distribution of the longer side of lesion boxes in the training set, in approximate native pixels. (d) Lesion burden per image (boxes per image, $\log_{10}$ scale, outliers omitted).

2020), which contains the labelled training portion of the APTOS 2019 Blindness Detection challenge (Asia Pacific Tele-Ophthalmology Society, 2019) after resizing to 224×224 and Gaussian-filter normalisation in the style of the Kaggle 2015 winning solution (Graham, 2015). It contains 3664 PNG images in five folders (No_DR 1805, Mild 370, Moderate 999, Severe 193, Proliferative_DR 295; CC0 licence). This set differs from DDR in several ways that make it a demanding test of robustness: it originates from a different population and camera set (Aravind Eye Hospital, India), has a different grade distribution (49% grade 0, 27% grade 2), has no lesion annotation and, most importantly, has a resolution of only 224×224 , which removes microaneurysms and most small haemorrhages (Fig. 4). We therefore *only* use it for zero-shot grading evaluation of models trained on DDR, and not for segmentation or detection. Because the images are already filtered, we apply the same filter to DDR inputs wherever a model is to be transferred (Section 3.3).

3.3. Preprocessing

All photographs are processed as follows. (i) *Border cropping*: the bounding box of pixels whose maximum channel exceeds 12 is cropped and the result is zero-padded to a square, which removes the black margins that occupy a variable portion of DDR frames. (ii) *Resizing*: area interpolation to 512×512 for the segmentation network and to 256×256 for the image embedding. Binary lesion masks are resized with area interpolation of the floating-point mask followed by a threshold of 0.15, which preserves sub-pixel

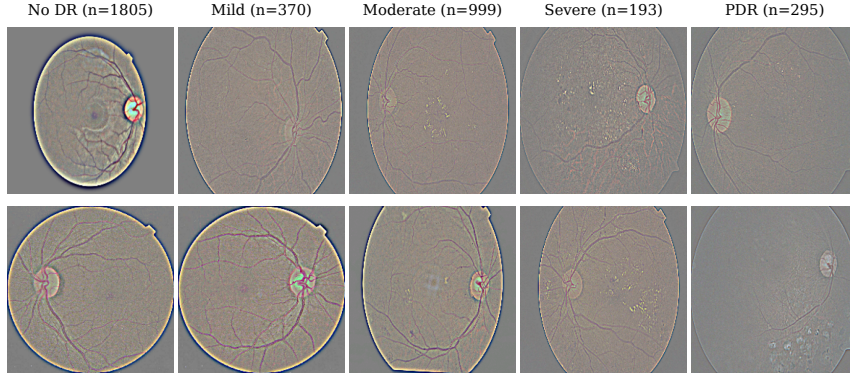


Figure 4: Random samples (two per grade) from the Gaussian-filtered 224×224 APTOS-derived Kaggle set. The Gaussian filter equalises illumination but also removes most colour information; note that some photographs are non-circular owing to the original acquisition.

microaneurysms that nearest-neighbour or threshold-at-0.5 resizing would erase; ground-truth boxes undergo exactly the same geometric transform. (iii) *Gaussian filtering* (optional, denoted “GF”): $I' = 4I - 4(G_\sigma * I) + 128$ with $\sigma = 0.045s$ (where s is the image side, matching the common $\sigma \approx 10$ at 224 px), followed by a gray fill outside the field of view. This is the same transform used to produce the Kaggle set, and we study its effect in an ablation (Section ??).

3.4. Data splits, leakage and ethics

We use the official DDR train/validation/test partition for every task. Model selection (early stopping, thresholds, temperature, conformal calibration) uses validation data only. A subtle point is that the lesion subset is *not* aligned with the grading partition: of the 383 lesion-training images, 275, 100 and 8 belong to the grading training, validation and test partitions, respectively; of the 149 lesion-validation images, 31 are in the grading validation and 118 in the grading test partition; and of the 225 lesion-test images, 152, 12 and 59 are in the grading training, validation and test partitions (Appendix Appendix A). Because the segmentation network is fitted on lesion-training images and tuned on lesion-validation images, any grading-test image that belongs to either set would give the lesion branch indirect access to test data. We therefore define a *leak-controlled grading test set* by removing these 126 images (all of which are gradable, grades 1–4), leaving 3633 gradable test images, and report *all* grading test results on this set;

results on the full official test partition are given as a sensitivity analysis in Appendix Appendix A. Lesion-test images are never used for fitting or selection of the segmentation network, so segmentation and detection results are unaffected. Both datasets are public and de-identified; no new patient data were collected, so institutional review was not required.

4. Method

4.1. Problem formulation and notation

Let $x \in \mathbb{R}^{H \times W \times 3}$ be a fundus photograph. We consider three coupled tasks on the same input. *Grading* asks for an ordinal grade $y \in \{0, 1, 2, 3, 4\}$ (Section 3.1). *Segmentation* asks for four binary masks $M_c \in \{0, 1\}^{H \times W}$ for lesion classes $c \in \mathcal{C} = \{\text{MA}, \text{HE}, \text{EX}, \text{SE}\}$. *Detection* asks for a set of scored boxes $\{(b_j, c_j, s_j)\}$. A segmentation network f_ϕ maps the image to lesion probability maps $P = f_\phi(x) \in [0, 1]^{4 \times H \times W}$; a grading model g_ψ maps the image and the lesion evidence derived from P to cumulative grade probabilities. Table 3 lists the main symbols.

Table 3: Main notation.

Symbol	Meaning
x, y	fundus image, ICDR grade $\in \{0, \dots, 4\}$
$\mathcal{C} = \{\text{MA}, \text{HE}, \text{EX}, \text{SE}\}$	lesion classes
$P = f_\phi(x) \in [0, 1]^{4 \times H \times W}$	lesion probability maps from the segmentation network
$\tilde{P} \in [0, 1]^{4 \times h \times w}$	$4 \times$ max-pooled maps ($h = w = 128$)
$\mathcal{Z} = \{z_1, \dots, z_6\}, \Omega_z$	retinal zones (4 quadrants, central disc, whole field) and their pixel sets
$n_{c,z}^{(\tau)}, \pi_{c,z}, \mu_{c,z}$	soft counts at threshold τ , number of peaks, maximum probability
$e_{c,z} \in \mathbb{R}^5, Z \in \mathbb{R}^{4 \times 6 \times 5}$	evidence vector of (c, z) and the full evidence tensor
$\varphi(x) \in \mathbb{R}^D$	frozen image embedding ($D = 1152$)
$s_k, \hat{F}_k = \sigma(s_k)$	cumulative logit and probability for the event $y \geq k, k = 1, \dots, 4$
$R_k(Z; \theta) \in [0, 1]$	soft-logic rule score: necessary condition for $y \geq k$
σ_y, λ	soft-label width, rule-loss weight
$\mathcal{C}_\alpha(x)$	conformal prediction set at miscoverage level α

4.2. Overview

Fig. 5 shows the overall pipeline. The segmentation network is trained on the pixel-annotated DDR subset and applied to every image. Its probability

maps are summarised by the lesion-evidence bottleneck into a tensor Z of soft, zone-wise lesion statistics (Section 4.5). The grading head combines a frozen embedding of the same image with the evidence tokens through a transformer (Section 4.6) and outputs cumulative ordinal probabilities (Section 4.7). A rule module computes ICDR-style necessary-condition scores $R_k(Z)$ and penalises predictions that exceed them (Section 4.8). Probabilities are temperature-scaled and converted into conformal prediction sets (Section 4.9). Detection boxes are obtained from the same probability maps P (Section 4.4).

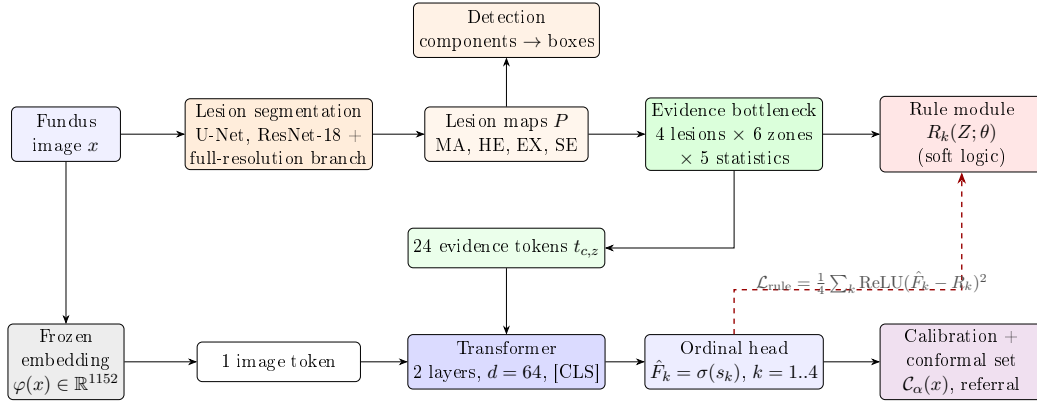


Figure 5: Overview of LesionRule. A segmentation network produces lesion maps; a differentiable bottleneck turns them into zone-wise evidence tokens, which a transformer combines with a frozen image embedding to predict cumulative ordinal probabilities. A soft-logic rule module (dashed) penalises grades that the lesion evidence cannot justify. Probabilities are calibrated and wrapped into conformal sets. The same lesion maps yield detection boxes.

4.3. Lesion segmentation network

Architecture.. The segmenter f_ϕ is a U-Net (Ronneberger et al., 2015) with a ResNet-18 encoder (He et al., 2016) initialised from ImageNet weights (as distributed in `timm` (Wightman, 2019)). The encoder produces features at strides 2, 4, 8, 16 and 32; the decoder upsamples bilinearly and fuses encoder features by concatenation followed by two 3×3 convolution–BN–ReLU blocks (widths 128, 96, 64, 48). Because micro-aneurysms are only a few pixels wide, stride-2 features are not sufficient, and we add a shallow *full-resolution branch* (two 3×3 conv–BN–ReLU blocks with 16 channels applied directly to the input) whose output is concatenated with the upsampled decoder output

before a final 3×3 block and a 1×1 convolution to four logits. The network has 11.8 M parameters.

Objective.. Let $p_c = \sigma(\ell_c)$ be the predicted probability map and y_c the binary mask of class c . We compare three objectives:

$$\mathcal{L}_{\text{bce}} = \frac{1}{4} \sum_c \text{BCE}(p_c, y_c), \quad (1)$$

$$\mathcal{L}_{\text{bce+dice}} = \frac{1}{4} \sum_c [\text{BCE}_w(p_c, y_c) + \mathcal{L}_{\text{Dice}}(p_c, y_c)], \quad (2)$$

$$\mathcal{L}_{\text{seg}} = \frac{1}{4} \sum_c [\text{BCE}_w(p_c, y_c) + 2 \text{FTL}_c], \quad (3)$$

where BCE_w weights positive pixels by $w = (20, 6, 6, 8)$ for (MA, HE, EX, SE), reflecting that the smaller and rarer the lesion class, the stronger the class imbalance. $\mathcal{L}_{\text{Dice}} = 1 - \frac{2 \sum p y + 1}{\sum p + \sum y + 1}$, and the focal Tversky loss (Abraham and Khan, 2019; Salehi et al., 2017) is

$$\text{FTL}_c = \left(1 - \text{TI}_c\right)^\gamma, \quad \text{TI}_c = \frac{\text{TP}_c + 1}{\text{TP}_c + \alpha \text{FP}_c + \beta \text{FN}_c + 1}, \quad (4)$$

with $\text{TP}_c = \sum p_c y_c$, $\text{FP}_c = \sum p_c (1 - y_c)$, $\text{FN}_c = \sum (1 - p_c) y_c$ computed over the mini-batch, $\alpha = 0.3$, $\beta = 0.7$ and $\gamma = 0.75$. Since $\beta > \alpha$, missed lesion pixels cost more than spurious ones, which is the desirable bias for screening where recall of small lesions dominates; $\gamma < 1$ gives larger gradients to well-classified classes than the plain Tversky loss, and keeps the loss sensitive when overlap is already moderate. $\mathcal{L}_{\text{seg}}$ is our proposed objective; $\mathcal{L}_{\text{bce}}$ and $\mathcal{L}_{\text{bce+dice}}$ serve as ablation baselines.

Lesion-centred sampling and optimisation.. Positive pixels are less than 1% of all pixels. Each training crop (256×256) is, with probability 0.7, positioned around a pixel of a randomly chosen lesion class in an image containing that class (the offset of the lesion inside the crop is random), and with probability 0.3 sampled uniformly. We use random flips, 90° rotations and brightness/contrast jitter. Optimisation uses AdamW (Loshchilov and Hutter, 2019) with weight decay 10^{-4} , one-cycle learning-rate schedule (peak 10^{-3} , 15% warm-up), batches of 8 crops, 48 iterations per epoch, 30 epochs and bfloat16 autocast on CPU (Alg. 1). The checkpoint with the highest mean AUPR on the lesion-validation split (evaluated every five epochs at the full 512×512 resolution) is retained.

Algorithm 1 Lesion-centred training of the segmentation network.

Require: lesion-train images $\{(x_i, Y_i)\}$, epochs E , iterations I , crop size q , probability $\rho = 0.7$

```
1: for  $e = 1, \dots, E$  do
2:   for  $t = 1, \dots, I$  do
3:     for  $b = 1, \dots, 8$  do
4:       draw  $u \sim \mathcal{U}(0, 1)$ 
5:       if  $u < \rho$  then
6:         draw class  $c$ , image  $i$  with  $Y_{i,c} \neq 0$ , lesion pixel  $r$ ; place
        crop containing  $r$  at a random offset
7:       else
8:         draw image  $i$  and crop position uniformly
9:       end if
10:      augment (flip, rot90°, brightness/contrast) crop and mask
11:    end for
12:     $\ell \leftarrow f_\phi(\text{batch})$  (bf16);  $\mathcal{L} \leftarrow \mathcal{L}_{\text{seg}}(\ell, \text{masks})$  with Eq. (4)
13:    update  $\phi$  with AdamW and one-cycle schedule
14:  end for
15:  if  $e \bmod 5 = 0$  then
16:    evaluate mean AUPR on lesion-valid at 5122; keep best  $\phi$ 
17:  end if
18: end for
```

4.4. Detection from segmentation maps

For a class c and image, the probability map P_c is thresholded at a seed threshold $\vartheta_c = \max(0.6t_c, 0.078)$, where t_c is the Dice-optimal threshold on the validation set. Each 8-connected component K becomes a box b_K (its tight bounding rectangle) with score $s_K = \frac{1}{|K|} \sum_{u \in K} P_c(u)$. Boxes with lower seed thresholds have higher recall but also more false positives; their scores order them for precision–recall and FROC analysis (Alg. 2). No extra network is trained: segmentation and detection share the same weights, and we evaluate the proposals directly against the DDR ground-truth boxes.

4.5. Lesion-evidence bottleneck

The grading head does not see P directly; it sees a compact, interpretable summary computed as follows.

Algorithm 2 Segmentation-guided lesion proposals and evaluation.

Require: probability maps $P \in [0, 1]^{4 \times H \times W}$, thresholds t_c (validation Dice-optimal)

- 1: **for** each class $c \in \mathcal{C}$ **do**
- 2: $\vartheta_c \leftarrow \max(0.6 t_c, 0.078)$; $B_c \leftarrow \emptyset$
- 3: find 8-connected components $\{K\}$ of $\{u : P_c(u) \geq \vartheta_c\}$
- 4: **for** each component K **do**
- 5: $b_K \leftarrow$ bounding box of K ; $s_K \leftarrow \text{mean}_{u \in K} P_c(u)$; add (b_K, s_K) to B_c
- 6: **end for**
- 7: **end for**
- 8: **evaluate:** sort B_c by score; greedy one-to-one matching to unique ground-truth boxes with $\text{IoU} \geq \kappa$ (AP_κ , $\kappa \in \{0.3, 0.5\}$); lesion-level FROC: a box is a hit if its centre lies inside a ground-truth box

Zones. Let $h = w = 128$ and let $\tilde{P} = \text{maxpool}_4(P) \in [0, 1]^{4 \times h \times w}$. Max-pooling (rather than average pooling) keeps the peak of a one-pixel microaneurysm. We define six zones with binary masks Ω_z on the $h \times w$ grid, restricted to the circular field of view of radius $0.5h - 2$: four quadrants (split by the horizontal and vertical lines through the image centre), a central disc of radius $0.35 \cdot (h/2)$ approximating the macular region, and the whole field. Because the cropped photographs are approximately centred on the optic disc-macula axis, quadrants are a pragmatic stand-in for the ETDRS/ICDR quadrant definition; we do not claim anatomical registration.

Soft counts. Counting connected components is not differentiable and is unstable under thresholding. Instead, for threshold levels $\tau \in \{0.25, 0.5, 0.75\}$ we compute

$$n_{c,z}^{(\tau)} = \sum_{u \in \Omega_z} \sigma\left(\frac{\tilde{P}_c(u) - \tau}{T}\right), \quad T = 0.05, \quad (5)$$

which counts the number of pooled cells in zone z that contain lesion class c with confidence about τ . Two further statistics are the number of local peaks $\pi_{c,z} = \sum_{u \in \Omega_z} \mathbb{1}[\tilde{P}_c(u) = \max_{u' \in \mathcal{N}_3(u)} \tilde{P}_c(u') \wedge \tilde{P}_c(u) > 0.5]$ (a proxy for the lesion count) and the maximum probability $\mu_{c,z} = \max_{u \in \Omega_z} \tilde{P}_c(u)$. The evidence vector is

$$e_{c,z} = \log(1 + [n_{c,z}^{(0.25)}, n_{c,z}^{(0.5)}, n_{c,z}^{(0.75)}, \pi_{c,z}, 10 \mu_{c,z}]) \in \mathbb{R}^5, \quad (6)$$

and the tensor $Z = (e_{c,z}) \in \mathbb{R}^{4 \times 6 \times 5}$ (120 numbers) is the *only* way in which lesion information enters the grade (Alg. 3). The logarithm compresses the dynamic range between a handful of microaneurysms and several hundred exudate cells. Since Eq. (5) is smooth in $\tilde{P}$, the bottleneck is differentiable in principle, which would allow end-to-end fine-tuning of f_ϕ from grade supervision; in the present study, f_ϕ is frozen at this stage for compute reasons (Section 5).

Algorithm 3 Zone-wise soft lesion evidence.

Require: lesion probability maps $P \in [0, 1]^{4 \times H \times W}$, zone masks $\{\Omega_z\}_{z=1}^6$ on the 128^2 grid

- 1: $\tilde{P} \leftarrow \text{maxpool}_4(P)$
- 2: **for** $c \in \mathcal{C}$, $z = 1..6$ **do**
- 3: **for** $\tau \in \{0.25, 0.5, 0.75\}$ **do**
- 4: $n_{c,z}^{(\tau)} \leftarrow \sum_{u \in \Omega_z} \sigma((\tilde{P}_c(u) - \tau)/T)$
- 5: **end for**
- 6: $\pi_{c,z} \leftarrow \sum_{u \in \Omega_z} \mathbb{I}[\tilde{P}_c(u) \text{ is a } 3 \times 3 \text{ local maximum, } \tilde{P}_c(u) > 0.5]$
- 7: $\mu_{c,z} \leftarrow \max_{u \in \Omega_z} \tilde{P}_c(u)$
- 8: $e_{c,z} \leftarrow \log(1 + [n^{(.25)}, n^{(.5)}, n^{(.75)}, \pi, 10\mu])$
- 9: **end for**
- 10: **return** $Z = (e_{c,z})$

4.6. Evidence-token grade head

Each (c, z) pair becomes a token $t_{c,z} = W_e e_{c,z} + a_c + b_z \in \mathbb{R}^d$ with $d = 64$, where a_c and b_z are learned class and zone embeddings. The frozen image embedding $\varphi(x) \in \mathbb{R}^{1152}$ (concatenated global averages of stages 3 and 4 of a ConvNeXt-T (Liu et al., 2022) applied to the 256×256 filtered image, standardised with training-set statistics) becomes one image token $t_0 = W_i \text{Dropout}(\varphi(x))$. A learned [CLS] token and the $1 + 24$ tokens enter a two-layer pre-LN transformer encoder (four heads, feed-forward width 128, dropout 0.1). The output of the [CLS] token passes through LayerNorm and a linear layer to produce four cumulative logits $s_1, \dots, s_4$. Attention weights from the [CLS] token to the evidence tokens provide a direct, inspectable account of which lesion class and zone drove a prediction. The head has 0.14 M parameters, so that the whole grading stage can be trained in seconds on a CPU.

4.7. Ordinal cumulative-link objective with soft targets

Following the multiple-output formulation of ordinal regression (Cao et al., 2020; Niu et al., 2016), we model $\hat{F}_k = P(y \geq k \mid x) = \sigma(s_k)$ for $k = 1, \dots, 4$. To model adjacent-grade disagreement we replace the one-hot target by a discretised Gaussian over grades,

$$q_j(y) = \frac{\exp(-(j-y)^2/2\sigma_y^2)}{\sum_{j'=0}^4 \exp(-(j'-y)^2/2\sigma_y^2)}, \quad F_k^* = \sum_{j \geq k} q_j(y), \quad (7)$$

and minimise the binary cross-entropy between F_k^* and $\hat{F}_k$:

$$\mathcal{L}_{\text{ord}} = -\frac{1}{4} \sum_{k=1}^4 \left[F_k^* \log \hat{F}_k + (1 - F_k^*) \log(1 - \hat{F}_k) \right]. \quad (8)$$

With $\sigma_y = 0$ this reduces to the standard cumulative-link loss. At inference, $\hat{y} = \sum_k \mathbb{I}[\hat{F}_k > 0.5]$, and the class probabilities are obtained from the monotonised cumulative probabilities $\bar{F}_k = \min_{k' \leq k} \hat{F}_{k'}$ as $p_j = \bar{F}_j - \bar{F}_{j+1}$ (with $\bar{F}_0 = 1$, $\bar{F}_5 = 0$), floored at 10^{-6} and renormalised. In our experiments, $\sigma_y = 0.5$, which places about 0.79 of the target mass on the labelled grade and 0.10 on each neighbour.

4.8. Clinical-rule consistency

The ICDR scale describes grades through necessary conditions on the lesions that must be visible. We encode four such conditions as soft logic (Diligenti et al., 2017; Hu et al., 2016), using the product t-norm for conjunction ($a \wedge b = ab$) and its dual for disjunction ($a \vee b = 1 - (1 - a)(1 - b)$). Let $u_{c,z} = e_{c,z,2} = \log(1 + n_{c,z}^{(0.5)})$ and $a(v; \theta) = \sigma((v - \theta)/s)$ with $s = 0.35$. With learnable thresholds $\theta = (\theta_1, \dots, \theta_{10})$ and $z = \text{all the whole-field zone}$, the rule scores are

$$R_1 = \bigvee_{c \in C} a(u_{c,\text{all}}; \theta_c) \quad (\text{any lesion is visible: needed for r}) \quad (9)$$

$$R_2 = a(u_{\text{MA},\text{all}}; \theta_5) \vee a(u_{\text{HE},\text{all}}; \theta_2) \vee a(u_{\text{EX},\text{all}}; \theta_3) \vee a(u_{\text{SE},\text{all}}; \theta_4) \quad (\text{more than few microaneurysms}) \quad (10)$$

$$R_3 = \left(\bigwedge_{q=1}^4 a(u_{\text{HE},q}; \theta_{5+q}) \right) \vee a(u_{\text{SE},\text{all}}; \theta_{10}) \quad (\text{haemorrhages in four quadrants,}) \quad (11)$$

$$R_4 = R_2 \quad (\text{PDR is never lesion-free}) \quad (12)$$

where θ_c in Eq. (9) denotes thresholds $\theta_1, \dots, \theta_4$ for $c = \text{MA, HE, EX, SE}$. Equation (11) is a soft proxy for the “4” of the 4-2-1 rule; venous beading, IRMA and neovascularisation are not annotated in DDR, so R_3 is only a partial surrogate and R_4 imposes just a weak necessary condition. We stress that these rules are *necessary conditions* (“ $y \geq k \Rightarrow R_k$ ”), not sufficient ones: a grade may be lower than the lesions allow, but never higher.

Calibrating thresholds on labels.. Because the predicted lesion statistics are noisy, hand-set thresholds would be arbitrary. We therefore fit θ once on the training labels by minimising

$$\mathcal{L}_{\text{fit}}(\theta) = \sum_{k=1}^4 \left[\mathbb{E}_{i: y_i \geq k} (1 - R_k(Z_i; \theta)) + \frac{1}{2} \mathbb{E}_{i: y_i < k} R_k(Z_i; \theta) \right], \quad (13)$$

that is, the rule should hold for images whose grade is at least k (first term), and should not hold trivially for lesion-poor images (second term, which prevents $R_k \equiv 1$). The parameters are then frozen. The rules are thus *clinically inspired, data-calibrated templates*, not a re-implementation of the grading guideline.

Rule-consistency loss and violation rate.. The grading head is trained with

$$\mathcal{L} = \mathcal{L}_{\text{ord}} + \lambda \mathcal{L}_{\text{rule}}, \quad \mathcal{L}_{\text{rule}} = \frac{1}{4} \sum_{k=1}^4 \text{ReLU}(\hat{F}_k - R_k(Z; \theta))^2, \quad \lambda = 1, \quad (14)$$

so that the predicted probability of grade $\geq k$ is pushed below the rule score. As an evaluation measure independent of the training loss, we define the *rule-violation rate* of a model $\hat{y}(\cdot)$ on a test set as the fraction of images for which $\hat{y} \geq k$ and $R_k < 0.5$ for at least one $k \leq \hat{y}$ (Alg. 4); the same frozen rules are applied to *every* model, including baselines that never saw them.

4.9. Calibration and split-conformal referral

Temperature scaling.. On validation half A we fit a single temperature $T^* \in [0.5, 3]$ that minimises the negative log-likelihood of the class probabilities obtained from the logits s_k/T (Guo et al., 2017); this changes confidences but not predicted grades. Expected calibration error (ECE) is computed with 15 equal-width confidence bins.

Algorithm 4 Rule calibration, rule-consistent training and violation rate.

Require: training evidence $\{Z_i\}$, labels $\{y_i\}$, embeddings $\{\varphi_i\}$, σ_y , λ

- 1: **Calibrate rules:** initialise θ ; minimise $\mathcal{L}_{\text{fit}}(\theta)$ (Eq. 13) with Adam; freeze θ
- 2: **for** epoch = 1, . . . , 60 **do**
- 3: **for** mini-batch B **do**
- 4: $s \leftarrow g_\psi(\varphi, Z)$; $\hat{F} \leftarrow \sigma(s)$
- 5: $\mathcal{L} \leftarrow \mathcal{L}_{\text{ord}}(\hat{F}, F^*(y; \sigma_y)) + \lambda \cdot \frac{1}{4} \sum_k \text{ReLU}(\hat{F}_k - R_k(Z; \theta))^2$
- 6: update ψ (AdamW, cosine schedule)
- 7: **end for**
- 8: every 3 epochs: compute quadratic-weighted kappa on validation half A; keep best ψ
- 9: **end for**
- 10: **Violation rate** on test set: $\frac{1}{N} \sum_i \mathbb{I}[\exists k \leq \hat{y}_i : R_k(Z_i; \theta) < 0.5]$

Split-conformal sets.. Let $\hat{p}(x) \in \Delta^4$ denote the calibrated class probabilities. On validation half B (calibration set of size n , disjoint from model selection and temperature fitting), we define the nonconformity score $\varsigma_i = 1 - \hat{p}_{y_i}(x_i)$ and the threshold

$$\hat{q}_\alpha = \varsigma_{(\lceil (n+1)(1-\alpha) \rceil)}, \quad \mathcal{C}_\alpha(x) = \{j : 1 - \hat{p}_j(x) \leq \hat{q}_\alpha\}, \quad (15)$$

where $\varsigma_{(m)}$ is the m -th smallest score. If calibration and test data are exchangeable, $\Pr[y \in \mathcal{C}_\alpha(x)] \geq 1 - \alpha$ marginally (Angelopoulos and Bates, 2023; Vovk et al., 2005). Sets of size one are confident decisions; larger sets signal ambiguity. We define a *referral rule*: refer the patient if $\mathcal{C}_\alpha(x)$ contains any grade ≥ 2 (referable DR); because the guarantee concerns the true grade, the probability that a truly referable patient is *not* referred is at most α *marginally over the population*, but it need not hold within each class. We therefore also report the empirical class-conditional referral sensitivity (Alg. 5).

4.10. Image-quality gate

Under the six-class protocol, ungradable images (grade 5 in DDR) are detected by a separate binary classifier with the same MLP architecture as the baseline, trained on all training images with the target $\mathbb{I}[y = 5]$. Its decision threshold is chosen on validation data to maximise the macro-F1 of the six-class prediction; an image flagged as ungradable receives class 5, otherwise

Algorithm 5 Calibration and split-conformal referral.

Require: trained head, validation halves A and B, test image x , level α

- 1: $T^* \leftarrow \arg \min_T \text{NLL}(\text{half A}; T)$
 - 2: for $i \in \text{half B}$: $\varsigma_i \leftarrow 1 - \hat{p}_{y_i}(x_i; T^*)$
 - 3: $\hat{q}_\alpha \leftarrow$ the $\lceil (n+1)(1-\alpha) \rceil$ -th smallest ς_i
 - 4: $\mathcal{C}_\alpha(x) \leftarrow \{j : 1 - \hat{p}_j(x; T^*) \leq \hat{q}_\alpha\}$
 - 5: **refer** iff $\max \mathcal{C}_\alpha(x) \geq 2$
-

the ordinal grade is output. The gate is deliberately kept independent of the ordinal head, so that the five-class results are not affected by it.

4.11. Complexity

The segmentation network (11.8 M parameters) is the dominant cost: on four CPU cores with bfloat16 it processes about 8 images per second at 512×512 (Section 5). Evidence extraction (Alg. 3) costs $\mathcal{O}(4 \cdot 6 \cdot hw)$ per image and is negligible. The grading head has 0.14 M parameters, and each ordinal prediction needs 26 tokens, i.e. $\mathcal{O}(26^2 d)$ attention operations. Conformal calibration needs a sort of n scores. The method thus adds under 1% of parameters and well under 1% of the compute of the segmentation network on top of a backbone embedding.

5. Experimental setup

5.1. Compute environment and its consequences

All experiments were run on a single cloud virtual machine with four CPU cores of an Intel Xeon (2.1 GHz, AVX-512 and AMX bfloat16 instructions), 15 GB of memory and no GPU. PyTorch (Paszke et al., 2019) with bfloat16 autocast and channels-last tensors gave a 3–4 $\times$ speed-up over float32 on this hardware, which made the following design feasible: the segmentation network trains in about 23 minutes (30 epochs) and processes about eight 512×512 images per second at inference. Image embeddings come from a *frozen* ImageNet-pretrained backbone, so that the grading stage trains in seconds and can be repeated over many seeds. This design imposes limitations that readers should keep in mind: (i) the segmentation network is trained at 512×512 rather than at native resolution (about 1950×1950), which shrinks microaneurysms to one to three pixels; (ii) it is trained on the 383 lesion-training images only; (iii) the image backbone is not fine-tuned on DR;

(iv) segmentation results come from a single training run per configuration (seed 0), while grading-stage results average five seeds; and (v) heavy detectors (Faster R-CNN, DETR) and large transformer backbones could not be trained within the budget, so detection is evaluated through segmentation-derived proposals against a classical image-processing baseline and across segmentation losses.

5.2. Implementation details

Segmentation. Architecture, losses, sampling and optimisation are as described in Section 4.3. Variants trained: (a) $\mathcal{L}_{\text{bce}}$, (b) $\mathcal{L}_{\text{bce}+\text{dice}}$, (c) $\mathcal{L}_{\text{seg}}$ (focal Tversky) on raw colour input, and (d) $\mathcal{L}_{\text{seg}}$ on Gaussian-filtered input (“GF”; Section 3.3). All use the same seed, crop sampler and schedule. Thresholds for Dice/IoU are chosen on lesion-validation data per class.

Image embedding. ConvNeXt-T (ImageNet-1k weights `convnext_tiny.fb_in1k` from `timm`) is applied to 256×256 Gaussian-filtered images; global averages of stage 3 (384 channels) and stage 4 (768 channels) are concatenated to a 1152-dimensional vector and standardised using training-set statistics. We chose this backbone and resolution a priori for throughput reasons; no alternative was evaluated on the test set.

Grading heads. All heads are trained with AdamW (weight decay 0.01), cosine schedule over 60 epochs, mini-batches of 128, with learning rate 10^{-3} for MLPs and 5×10^{-4} for the token head; the epoch with the highest quadratic-weighted kappa on validation half A (checked every three epochs) is retained. Validation is split into two halves with a fixed random permutation: half A (1251 images) for model selection and temperature, half B (1252 images) for conformal calibration. Each configuration is trained with five seeds (0–4); the data split is the same for all.

5.3. Compared methods

Table 4 lists the grading models. B1–B4 share the same embedding and (where applicable) the same lesion evidence as the proposed models, so that every difference is attributable to *how* lesion information is used.

For segmentation, the four U-Net variants of Section 4.3 are compared. For detection we compare segmentation-derived proposals from each variant and a classical baseline: top-hat filtering of the green channel followed by threshold and connected components (dark top-hat for MA and HE,

Table 4: Grading models compared. All use the same frozen ConvNeXt-T embedding $\varphi(x)$; “evidence” means the tensor Z derived from the segmentation maps of the same segmentation network.

Model	Description	Evidence	Ordinal	Soft	Rule
B1	MLP on $\varphi(x)$, 5-way cross-entropy	–	–	–	–
B2	MLP on $\varphi(x)$, cumulative-link loss (Eq. 8, $\sigma_y = 0$)	–	✓	–	–
B3	MLP on the flattened evidence Z only	✓	✓	–	–
B4	MLP on concatenation $[\varphi(x), Z]$ (naive fusion)	✓	✓	–	–
T0	Evidence-token transformer, $\sigma_y = 0$, $\lambda = 0$	✓	✓	–	–
T1	T0 with soft targets ($\sigma_y = 0.5$)	✓	✓	✓	–
T2	T0 with rule loss ($\lambda = 1$)	✓	✓	–	✓
LesionRule	transformer, $\sigma_y = 0.5$, $\lambda = 1$	✓	✓	✓	✓
A1	LesionRule without evidence tokens in the head (image token only; rule loss still reads Z)	(rule only)	✓	✓	✓

bright top-hat for EX and SE), with the threshold tuned per class on lesion-validation data. We do not compare against published numbers of other systems because their splits, resolutions and training budgets differ.

5.4. Evaluation metrics

Grading.. The primary metric is the quadratic-weighted Cohen’s kappa (QWK) (Cohen, 1968), the standard agreement measure for ordinal DR grades. We also report accuracy, macro-averaged F1, the area under the ROC curve (AUC) for referable DR ($y \geq 2$) computed from $P(y \geq 2)$, expected calibration error (ECE, 15 bins), and the rule-violation rate (Section 4.8). Conformal prediction is evaluated at $\alpha \in \{0.05, 0.1, 0.2\}$ by empirical coverage, mean set size, singleton fraction and referral sensitivity/specificity. All grading metrics are computed on the leak-controlled test set (Section 3) unless stated.

Segmentation.. The area under the precision–recall curve (AUPR) of the pixel probabilities, computed exactly after quantising probabilities to 512 levels, is the primary metric because it is threshold-free and appropriate for extreme class imbalance (Saito and Rehmsmeier, 2015). We additionally report Dice (Dice, 1945) and intersection-over-union (IoU) at the per-class validation-optimal threshold, accumulated over all test pixels.

Detection.. Average precision at IoU 0.5 and 0.3 (all-point interpolation (Padilla et al., 2021)) and lesion-level FROC sensitivity at 1, 2, 4, 8 and 16 false positives per image, where a prediction is a hit if its centre lies in a ground-truth box. The relaxed IoU of 0.3 is reported because, for a box a few pixels wide, one pixel of displacement lowers IoU dramatically.

5.5. Statistical analysis

Grading results are reported as mean $\pm$ standard deviation over five seeds. For the main comparisons we additionally report 95% confidence intervals obtained by a non-parametric bootstrap of the test images (Efron and Tibshirani, 1993) (2000 resamples) applied to the seed-averaged class probabilities, and paired bootstrap differences in QWK between models, which respect the pairing of test images. Differences are deemed significant when the 95% interval of the paired difference excludes zero; no multiplicity correction is applied to the exploratory ablation table, and we say so where relevant. Segmentation metrics are reported from a single run per configuration; we therefore do not make significance claims among segmentation variants and we compare them descriptively.

5.6. Reproducibility

All scripts (data extraction, training, evaluation, figure and table generation) and intermediate artefacts (cached lesion maps, embeddings, per-seed probabilities) are released with the paper. Every number in the tables is generated by scripts from stored result files; there is no manual transcription. Random seeds are fixed. The code reads the DDR archive directly from its split zip volumes, so no re-packaging is needed. TODO TODO TODO

Declarations

CRediT authorship contribution statement.. [To be completed by the authors.] Conceptualization, Methodology, Software, Validation, Formal anal-

ysis, Investigation, Data curation, Writing – original draft, Writing – review & editing, Visualization, Supervision.

Declaration of competing interest.. [To be completed by the authors.] The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Funding.. [To be completed by the authors.]

Data and code availability.. The DDR dataset is available from its authors at <https://github.com/nkicsl/DDR-dataset> (links to Google Drive and Baidu Drive; CC BY-NC-SA 4.0). The Gaussian-filtered Kaggle dataset is available at <https://www.kaggle.com/datasets/sovittrath/diabetic-retinopathy-224x224-g> (CC0). Code, configuration files and result files that generate every table and figure in this paper are available in the repository <https://github.com/deep1789/Diabetic-Retinopathy>.

Declaration of generative AI and AI-assisted technologies in the writing process.. During the preparation of this work an AI assistant (Claude, Anthropic) was used to write code, run experiments and draft the manuscript text. The authors must review and edit all content and take full responsibility for the publication before submission. [Authors to confirm and adapt this statement according to the journal’s policy.]

Ethics.. This study uses only publicly available, de-identified images; no new patient data were collected.

Appendix A. Overlap between lesion and grading partitions, and sensitivity analysis

Table A.5 shows how the 757 lesion-annotated images are distributed over the grading partitions, computed by matching file names. The two partitions are not aligned. The 126 grading-test images that also occur in the lesion-training (8) or lesion-validation (118) subsets are removed from the grading test set in all main experiments (leak-controlled test set, 3633 gradable images). Table ?? compares the main grading metrics on the leak-controlled and on the full official test partition (3759 gradable images).

Table A.5: Number of lesion-subset images (rows) found in each grading partition (columns), by file name.

Lesion subset	Grading train	Grading valid	Grading test	Total
Lesion train (383)	275	100	8	383
Lesion valid (149)	0	31	118	149
Lesion test (225)	152	12	59	223*

* Two lesion-test images are not listed in any grading partition.

Appendix B. Hyper-parameters

Table B.6: Hyper-parameters. Values were fixed a priori or tuned on validation data only.

Component	Parameter	Value
Segmentation	input / crop	512 ² full image / 256 ² crops
	optimiser, schedule	AdamW, wd 10 ⁻⁴ , one-cycle (max 10 ⁻³ , 15% warm-
	batch, iterations × epochs	8 crops, 48 × 30
	positive weights (MA,HE,EX,SE)	20, 6, 6, 8
	focal Tversky	$\alpha = 0.3$, $\beta = 0.7$, $\gamma = 0.75$, weight 2
	lesion-centred crop probability	0.7
	augmentation	flips, 90° rotations, contrast × [0.8, 1.25], brightness ±
Evidence	grid, zones	128 ² , 6 (4 quadrants, central disc $r = 0.35R$, whole f
	soft-count thresholds, temperature	$\tau \in \{0.25, 0.5, 0.75\}$, $T = 0.05$
	peak rule	3 × 3 local maximum, $\tilde{P} > 0.5$
Grade head	token width, layers, heads, FFN	64, 2, 4, 128
	optimiser	AdamW, wd 0.01, cosine, 60 epochs, batch 128
	learning rate	5×10^{-4} (transformer), 10 ⁻³ (MLPs)
	soft-label width σ_y	0.5
	rule weight λ , rule slope s	1, 0.35
Rules	fitting	Adam (lr 0.05), 400 steps, negative weight 0.5
Calibration	temperature grid	26 values in [0.5, 3]
Conformal	α	0.05, 0.1, 0.2
Detection	seed threshold	$\max(0.6 t_c, 0.078)$

Appendix C. Properties of the building blocks

Proposition 1 (Soft counts). *For every zone z , $0 \leq n_{c,z}^{(\tau)} \leq |\Omega_z|$; $n_{c,z}^{(\tau)}$ is non-increasing in τ and non-decreasing in every entry of $\tilde{P}_c$; $|\partial n_{c,z}^{(\tau)} / \partial \tilde{P}_c(u)| \leq$*

$1/(4T)$; and as $T \rightarrow 0$ it converges, for all u with $\tilde{P}_c(u) \neq \tau$, to the hard count $|\{u \in \Omega_z : \tilde{P}_c(u) > \tau\}|$.

Proof. Each summand $\sigma((\tilde{P}_c(u) - \tau)/T)$ lies in $(0, 1)$, is decreasing in τ and increasing in $\tilde{P}_c(u)$, and has derivative $\sigma'(\cdot)/T \leq 1/(4T)$ because $\sigma' \leq 1/4$. Summing over $|\Omega_z|$ terms gives the bounds. As $T \rightarrow 0$, $\sigma((v - \tau)/T) \rightarrow \mathbb{I}[v > \tau]$ for $v \neq \tau$. $\square$

Proposition 2 (Monotone rules and the rule loss). *Let $R_k(Z; \theta)$ be defined by Eqs. (9)–(12). (i) $R_k \in (0, 1)$ and R_k is non-decreasing in every entry $u_{c,z}$ of the evidence, so additional lesion evidence can only relax a rule. (ii) For $\hat{F} \in [0, 1]^4$, $\mathcal{L}_{\text{rule}}(\hat{F}; R) = 0$ if and only if $\hat{F}_k \leq R_k$ for all k . (iii) $\mathcal{L}_{\text{rule}}$ is continuously differentiable in $\hat{F}$ with gradient $\frac{1}{2} \text{ReLU}(\hat{F}_k - R_k)$, which vanishes exactly on consistent predictions, so consistent predictions are not penalised.*

Proof. (i) $a(v; \theta) = \sigma((v - \theta)/s)$ is increasing in v with values in $(0, 1)$. Under the product t-norm, $\prod_i a_i$ is increasing in each a_i , and the dual $1 - \prod_i (1 - a_i)$ is increasing as well; compositions of increasing maps are increasing and remain in $(0, 1)$. (ii) A sum of squares of non-negative terms is zero exactly if every term vanishes. (iii) $\frac{d}{dx} \text{ReLU}(x)^2 = 2 \text{ReLU}(x)$, which is continuous and zero for $x \leq 0$; the factor $\frac{1}{4} \cdot 2 = \frac{1}{2}$ follows. $\square$

Proposition 3 (Ordinal soft targets). *For any $\sigma_y > 0$ the targets $F_k^* = \sum_{j \geq k} q_j(y)$ in Eq. (7) satisfy $1 \geq F_1^* \geq \dots \geq F_4^* \geq 0$, and $F_k^* \rightarrow \mathbb{I}[y \geq k]$ as $\sigma_y \rightarrow 0$. Consequently the cumulative-link loss with soft targets has the same minimiser as the hard-label loss when $\sigma_y \rightarrow 0$, and for $\sigma_y > 0$ its population minimiser is $\hat{F}_k = \mathbb{E}[F_k^* | x]$, a monotone sequence.*

Proof. $q_j \geq 0$ and $\sum_j q_j = 1$, so tail sums are non-increasing in k and lie in $[0, 1]$. As $\sigma_y \rightarrow 0$, $q_j(y) \rightarrow \mathbb{I}[j = y]$ and hence $F_k^* \rightarrow \mathbb{I}[y \geq k]$. For each k the binary cross-entropy $-F^* \log \hat{F} - (1 - F^*) \log(1 - \hat{F})$ with soft target F^* is minimised in expectation by $\hat{F} = \mathbb{E}[F^* | x]$; monotonicity in k follows from monotonicity of F_k^* pointwise. $\square$

Proposition 4 (Split-conformal coverage). *Let $(x_i, y_i)_{i=1}^n$ be calibration data and (x, y) a test pair, all exchangeable, and $\hat{q}_\alpha$ the $\lceil (n+1)(1-\alpha) \rceil$ -th smallest of the scores $s_i = 1 - \hat{p}_{y_i}(x_i)$ (set $\hat{q}_\alpha = \infty$ if the index exceeds n). Then $\Pr[y \in \mathcal{C}_\alpha(x)] \geq 1 - \alpha$, and if scores are almost surely distinct, $\Pr[y \in \mathcal{C}_\alpha(x)] \leq 1 - \alpha + 1/(n+1)$.*

Proof. Since $y \in \mathcal{C}_\alpha(x) \iff \varsigma_{n+1} := 1 - \hat{p}_y(x) \leq \hat{q}_\alpha$ and the $n + 1$ scores are exchangeable, the rank of ς_{n+1} among them is uniform on $\{1, \dots, n + 1\}$ when scores are distinct (and stochastically no smaller otherwise). The event $\varsigma_{n+1} \leq \hat{q}_\alpha$ is the event that its rank is at most $m = \lceil (n + 1)(1 - \alpha) \rceil$, which has probability $m/(n + 1) \geq 1 - \alpha$ and $\leq (1 - \alpha) + 1/(n + 1)$. $\square$

Note that the guarantee is marginal over the population (not conditional on the grade), requires the calibration half to be disjoint from model selection and temperature fitting (which is why validation is split into halves A and B), and presumes no distribution shift between calibration and test; Section ?? shows what happens under shift.

References

- Abraham, N., Khan, N.M., 2019. A novel focal Tversky loss function with improved attention U-Net for lesion segmentation, in: IEEE ISBI, pp. 683–687.
- Abràmoff, M.D., Lavin, P.T., Birch, M., Shah, N., Folk, J.C., 2018. Pivotal trial of an autonomous AI-based diagnostic system for detection of diabetic retinopathy in primary care offices. *npj Digital Medicine* 1, 39.
- Angelopoulos, A.N., Bates, S., 2023. Conformal prediction: A gentle introduction. *Foundations and Trends in Machine Learning* 16, 494–591.
- Asia Pacific Tele-Ophthalmology Society, 2019. APTOS 2019 blindness detection. Kaggle competition dataset.
- Cao, W., Mirjalili, V., Raschka, S., 2020. Rank consistent ordinal regression for neural networks with application to age estimation. *Pattern Recognition Letters* 140, 325–331.
- Chen, L.C., Zhu, Y., Papandreou, G., Schroff, F., Adam, H., 2018. Encoder-decoder with atrous separable convolution for semantic image segmentation, in: ECCV, pp. 801–818.
- Cohen, J., 1968. Weighted kappa: Nominal scale agreement with provision for scaled disagreement or partial credit. *Psychological Bulletin* 70, 213–220.

- Decencière, E., Cazuguel, G., Zhang, X., et al., 2013. TeleOphta: Machine learning and image processing methods for teleophthalmology. *IRBM* 34, 196–203.
- Dice, L.R., 1945. Measures of the amount of ecologic association between species. *Ecology* 26, 297–302.
- Diligenti, M., Gori, M., Saccà, C., 2017. Semantic-based regularization for learning and inference. *Artificial Intelligence* 244, 143–165.
- Dosovitskiy, A., Beyer, L., Kolesnikov, A., et al., 2021. An image is worth 16x16 words: Transformers for image recognition at scale, in: *ICLR*.
- Early Treatment Diabetic Retinopathy Study Research Group, 1991. Grading diabetic retinopathy from stereoscopic color fundus photographs – an extension of the modified Airlie House classification. ETDRS report number 10. *Ophthalmology* 98, 786–806.
- Efron, B., Tibshirani, R.J., 1993. *An Introduction to the Bootstrap*. Chapman & Hall.
- Foo, A., Hsu, W., Lee, M.L., Lim, G., Wong, T.Y., 2020. Multi-task learning for diabetic retinopathy grading and lesion segmentation, in: *AAAI*, pp. 13267–13272.
- Gal, Y., Ghahramani, Z., 2016. Dropout as a Bayesian approximation: Representing model uncertainty in deep learning, in: *ICML*, pp. 1050–1059.
- Graham, B., 2015. Kaggle diabetic retinopathy detection competition report. University of Warwick technical report.
- Gulshan, V., Peng, L., Coram, M., et al., 2016. Development and validation of a deep learning algorithm for detection of diabetic retinopathy in retinal fundus photographs. *JAMA* 316, 2402–2410.
- Guo, C., Pleiss, G., Sun, Y., Weinberger, K.Q., 2017. On calibration of modern neural networks, in: *ICML*, pp. 1321–1330.
- He, A., Li, T., Li, N., Wang, K., Fu, H., 2021. CABNet: Category attention block for imbalanced diabetic retinopathy grading. *IEEE Transactions on Medical Imaging* 40, 143–153.

- He, K., Zhang, X., Ren, S., Sun, J., 2016. Deep residual learning for image recognition, in: CVPR, pp. 770–778.
- Hu, Z., Ma, X., Liu, Z., Hovy, E., Xing, E., 2016. Harnessing deep neural networks with logic rules, in: ACL, pp. 2410–2420.
- Huang, Y., Lin, L., Cheng, P., Lyu, J., Tang, X., 2021. Lesion-based contrastive learning for diabetic retinopathy grading from fundus images, in: MICCAI, pp. 113–123.
- Ilse, M., Tomczak, J., Welling, M., 2018. Attention-based deep multiple instance learning, in: ICML, pp. 2127–2136.
- Li, T., Gao, Y., Wang, K., Guo, S., Liu, H., Kang, H., 2019. Diagnostic assessment of deep learning algorithms for diabetic retinopathy screening. *Information Sciences* 501, 511–522. doi:10.1016/j.ins.2019.06.011.
- Li, X., Hu, X., Yu, L., Zhu, L., Fu, C.W., Heng, P.A., 2020. CANet: Cross-disease attention network for joint diabetic retinopathy and diabetic macular edema grading. *IEEE Transactions on Medical Imaging* 39, 1483–1493.
- Lin, T.Y., Goyal, P., Girshick, R., He, K., Dollár, P., 2017. Focal loss for dense object detection, in: ICCV, pp. 2980–2988.
- Lin, Z., Guo, R., Wang, Y., et al., 2018. A framework for identifying diabetic retinopathy based on anti-noise detection and attention-based fusion, in: MICCAI, pp. 74–82.
- Liu, Z., Lin, Y., Cao, Y., et al., 2021. Swin transformer: Hierarchical vision transformer using shifted windows, in: ICCV, pp. 10012–10022.
- Liu, Z., Mao, H., Wu, C.Y., Feichtenhofer, C., Darrell, T., Xie, S., 2022. A ConvNet for the 2020s, in: CVPR, pp. 11976–11986.
- Loshchilov, I., Hutter, F., 2019. Decoupled weight decay regularization, in: ICLR.
- Niu, Z., Zhou, M., Wang, L., Gao, X., Hua, G., 2016. Ordinal regression with multiple output CNN for age estimation, in: CVPR, pp. 4920–4928.

- Oktay, O., Schlemper, J., Folgoc, L.L., et al., 2018. Attention U-Net: Learning where to look for the pancreas, in: Medical Imaging with Deep Learning (MIDL).
- Padilla, R., Passos, W.L., Dias, T.L.B., Netto, S.L., da Silva, E.A.B., 2021. A comparative analysis of object detection metrics with a companion open-source toolkit. *Electronics* 10, 279.
- Paszke, A., Gross, S., Massa, F., et al., 2019. PyTorch: An imperative style, high-performance deep learning library, in: NeurIPS, pp. 8024–8035.
- Playout, C., Duval, R., Cheriet, F., 2019. A novel weakly supervised multi-task architecture for retinal lesions segmentation on fundus images. *IEEE Transactions on Medical Imaging* 38, 2434–2444.
- Porwal, P., Pachade, S., Kokare, M., et al., 2020. IDRiD: Diabetic retinopathy – segmentation and grading challenge. *Medical Image Analysis* 59, 101561.
- Quelleg, G., Charrière, K., Boudi, Y., Cochener, B., Lamard, M., 2017. Deep image mining for diabetic retinopathy screening. *Medical Image Analysis* 39, 178–193.
- Rath, S.R., 2020. Diabetic retinopathy 224x224 Gaussian filtered. Kaggle dataset, derived from APTOS 2019 (CC0).
- Ren, S., He, K., Girshick, R., Sun, J., 2017. Faster R-CNN: Towards real-time object detection with region proposal networks. *IEEE Transactions on Pattern Analysis and Machine Intelligence* 39, 1137–1149.
- Ronneberger, O., Fischer, P., Brox, T., 2015. U-Net: Convolutional networks for biomedical image segmentation, in: MICCAI, pp. 234–241.
- Saeedi, P., Petersohn, I., Salpea, P., et al., 2019. Global and regional diabetes prevalence estimates for 2019 and projections for 2030 and 2045: Results from the IDF diabetes atlas, 9th edition. *Diabetes Research and Clinical Practice* 157, 107843.
- Saito, T., Rehmsmeier, M., 2015. The precision-recall plot is more informative than the ROC plot when evaluating binary classifiers on imbalanced datasets. *PLoS ONE* 10, e0118432.

- Salehi, S.S.M., Erdogmus, D., Gholipour, A., 2017. Tversky loss function for image segmentation using 3D fully convolutional deep networks, in: Machine Learning in Medical Imaging (MLMI), pp. 379–387.
- Shi, X., Cao, W., Raschka, S., 2023. Deep neural networks for rank-consistent ordinal regression based on conditional probabilities. *Pattern Analysis and Applications* 26, 941–955.
- Sun, R., Li, Y., Zhang, T., Mao, Z., Wu, F., Zhang, Y., 2021. Lesion-aware transformers for diabetic retinopathy grading, in: CVPR, pp. 10938–10947.
- Tan, M., Le, Q.V., 2019. EfficientNet: Rethinking model scaling for convolutional neural networks, in: ICML, pp. 6105–6114.
- Teo, Z.L., Tham, Y.C., Yu, M., et al., 2021. Global prevalence of diabetic retinopathy and projection of burden through 2045. *Ophthalmology* 128, 1580–1591.
- Tian, Z., Shen, C., Chen, H., He, T., 2019. FCOS: Fully convolutional one-stage object detection, in: ICCV, pp. 9627–9636.
- Ting, D.S.W., Cheung, C.Y.L., Lim, G., et al., 2017. Development and validation of a deep learning system for diabetic retinopathy and related eye diseases using retinal images from multiethnic populations with diabetes. *JAMA* 318, 2211–2223.
- Vovk, V., Gammerman, A., Shafer, G., 2005. Algorithmic Learning in a Random World. Springer.
- Wang, Z., Yin, Y., Shi, J., Fang, W., Li, H., Wang, X., 2017. Zoom-in-Net: Deep mining lesions for diabetic retinopathy detection, in: MICCAI, pp. 267–275.
- Wightman, R., 2019. PyTorch image models. GitHub repository, <https://github.com/huggingface/pytorch-image-models>.
- Wilkinson, C.P., Ferris, F.L., Klein, R.E., et al., 2003. Proposed international clinical diabetic retinopathy and diabetic macular edema disease severity scales. *Ophthalmology* 110, 1677–1682.

- Xu, J., Zhang, Z., Friedman, T., Liang, Y., Van den Broeck, G., 2018. A semantic loss function for deep learning with symbolic knowledge, in: ICML, pp. 5502–5511.
- Yang, Y., Li, T., Li, W., Wu, H., Fan, W., Zhang, W., 2017. Lesion detection and grading of diabetic retinopathy via two-stages deep convolutional neural networks, in: MICCAI, pp. 533–540.
- Zhou, Y., Chia, M.A., Wagner, S.K., et al., 2023. A foundation model for generalizable disease detection from retinal images. *Nature* 622, 156–163.
- Zhou, Y., He, X., Huang, L., Liu, L., Zhu, F., Cui, S., Shao, L., 2019. Collaborative learning of semi-supervised segmentation and classification for medical images, in: CVPR, pp. 2079–2088.
- Zhou, Y., Wang, B., Huang, L., Cui, S., Shao, L., 2021. A benchmark for studying diabetic retinopathy: Segmentation, grading, and transferability. *IEEE Transactions on Medical Imaging* 40, 818–828.
- Zhou, Z., Siddiquee, M.M.R., Tajbakhsh, N., Liang, J., 2020. UNet++: Redesigning skip connections to exploit multiscale features in image segmentation. *IEEE Transactions on Medical Imaging* 39, 1856–1867.